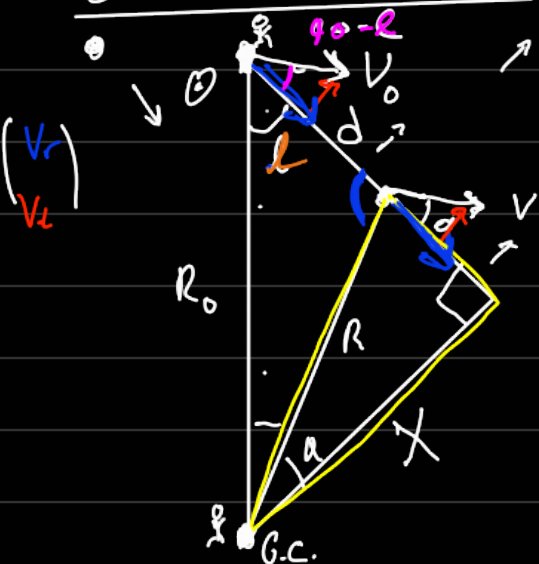


* Galaxies + velocities: (R, d, h)

$\begin{cases} V_r = A \sin(2\ell) \\ V_t = d(2\cos(2\ell) + B) \end{cases}$

(A, B)



$$\cos(\alpha) = \frac{x}{R}$$

$$\sin(\ell) = \frac{x}{R_0}$$

$$* (R_0 \sin(\ell) = R \cos(\alpha))$$

$\Rightarrow d \ll R$ or R_0 : *

$$* (R - R_0) \approx d \cos(\ell)$$

R_0

$$\begin{aligned} V_r &= V \cos(\alpha) + V_0 \cos(90 - \ell) \\ &= V \cos(\alpha) - V_0 \sin(\ell) \\ \cos(\alpha) &= \frac{R_0}{R} \sin(\ell) \end{aligned}$$

$$V_r = \left(V \frac{R_0}{R} - V_0 \right) \sin(\ell)$$

$$= R_0 \left(\frac{V}{R} - \frac{V_0}{R_0} \right) \sin(\ell)$$

$$= R_0 (\omega - \omega_0) \sin(\ell)$$

$$(V_r, V_t) \rightarrow (A, B)$$

$$V^2 = V_t^2 + V_r^2$$

* Nearby gal/ gas clouds:

$$\underline{A \approx 15.3}$$

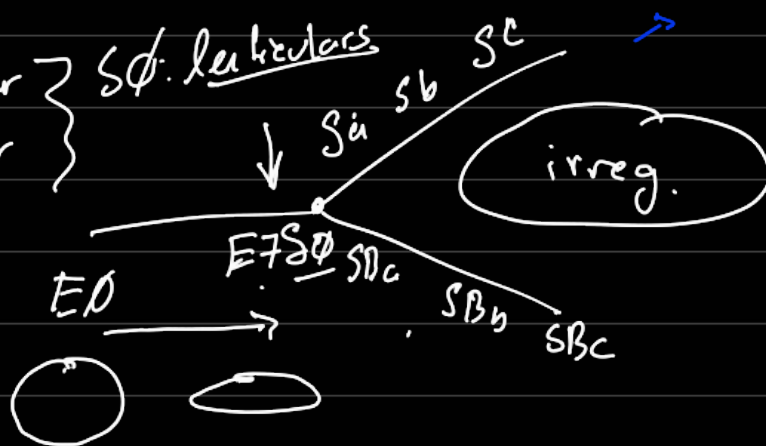
$$\underline{B \approx -11.9}$$

* Gava: * $V_t = 4.74 \text{ d (Kpc) } \mu (\text{milliwa. yr}^{-1})$ *

* Normal gal. classifications

$\begin{cases} E \text{ types} \\ S \text{ types} \end{cases}$
 • Dwarf

regular } $S\phi$. irregulars
 irregular }



$$E_n \leftarrow n = 10 \left(1 - \frac{b}{a} \right)$$

$$n=0: 0 = 1 - \frac{b}{a} \Rightarrow b=a$$

$$S_a \rightarrow S_c$$

• [B/D]



S_c / d



[B/D]

Assume B const:

D_c

$$\left(\frac{B}{D_a} \right) / \left(\frac{B}{D_c} \right) = \frac{D_c}{D_a} = \frac{1}{2} \quad D_a = 2 D_c$$

Surface brightness profiles:

Spirals: $I(R) = I_0 e^{-R/R_n}$

Ellipticals: $I(R) = I_c e^{-b \left(\frac{R}{R_c} \right)^{1/n} - 1}$

$n=4$ ellipticals, b_e

$n=1$: spirals, b_s

* Luminosity $\phi(L) dL$

[40] [41]: # of gal.

per unit vol
 w L between
 L and L+dL

$$L = \int_0^{2\pi} d\phi \int_0^{R'} R I(R) dR$$

$$R' = \infty$$

General: $L = \int_0^{2\pi} d\phi \int_0^{R'} R I_c e^{-b \left(\frac{R}{R_c} \right)^{1/n} - 1} dR$

$$L = 2\pi I_c \int_0^{\infty} R \frac{2\pi}{e^b} e^{-b \left(\frac{R}{R_c} \right)^{1/n}} dR$$

$$* = 2\pi I_c e^b \int_0^{\infty} R e^{-b \left(\frac{R}{R_c} \right)^{1/n}} dR$$

$$z = b \left(\frac{R}{R_h} \right)^{1/n} \quad dz = \frac{b}{n} \left(\frac{R}{R_h} \right)^{\frac{1}{n}-1} dR$$

$$dR = \frac{n}{b} z^{n-1} dz \quad \text{or} \quad = \frac{b}{n} \left(\frac{n}{R_h} \right)^{\frac{1}{n}-1} dR$$

$$\int \left(\frac{R}{b} \right)^n e^{-z} \left(\frac{n}{b} \right) z^{n-1} dz = \frac{b}{n} \left(\frac{n}{R_h} \right)^{\frac{1}{n}} \frac{1}{z^n} dz = \frac{1}{n} \frac{dz}{z^n}$$

$$= \frac{n}{b^{n+1}} \int z^{2n-1} e^{-z} dz$$

$$\Gamma(z+1) = \int dz z^z e^{-z} = z!$$

$$\frac{1}{b^{n+1}} 2n!$$

$$e^{-\frac{R}{R_h}}$$

$$L = 2\pi I_0 \int_0^R R e^{-\frac{R}{R_h}} dR \quad a = -\frac{1}{R_h}$$

$$\int_0^R x e^{ax} dx = \left[\frac{e^{ax}}{a^2} (ax - 1) \right]_0^R$$

$$= e^{-R/R_h} R_h^2 \left(-\frac{R}{R_h} - 1 \right) - \underline{1 R_h^2 (-1)}$$

$$= R_h^2 - e^{-R/R_h} R_h^2 \left(\frac{R}{R_h} + 1 \right)$$

$$L = 2\pi I_0 R_h^2 \left(1 - e^{-R/R_h} \left(\frac{R}{R_h} + 1 \right) \right)$$

$$\lim_{R \rightarrow \infty} \left(e^{-R/R_h} \left(\frac{R}{R_h} + 1 \right) \right) = 0$$

$$L_{tot} = 2\pi I_0 R_h^2$$

$e^{-\infty} \rightarrow 0$
much faster

